Authentication Module – Technical Documentation

Project: TaskFlow – Intelligent Task Allocation & Tracking System
Module: Authentication · **Child Modules:** Login, Account Recovery, MFA Security Setup, Sessions and Tokens
Version: 1.0 · **Last Updated:** 31-Oct-2026
Backend: ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core)
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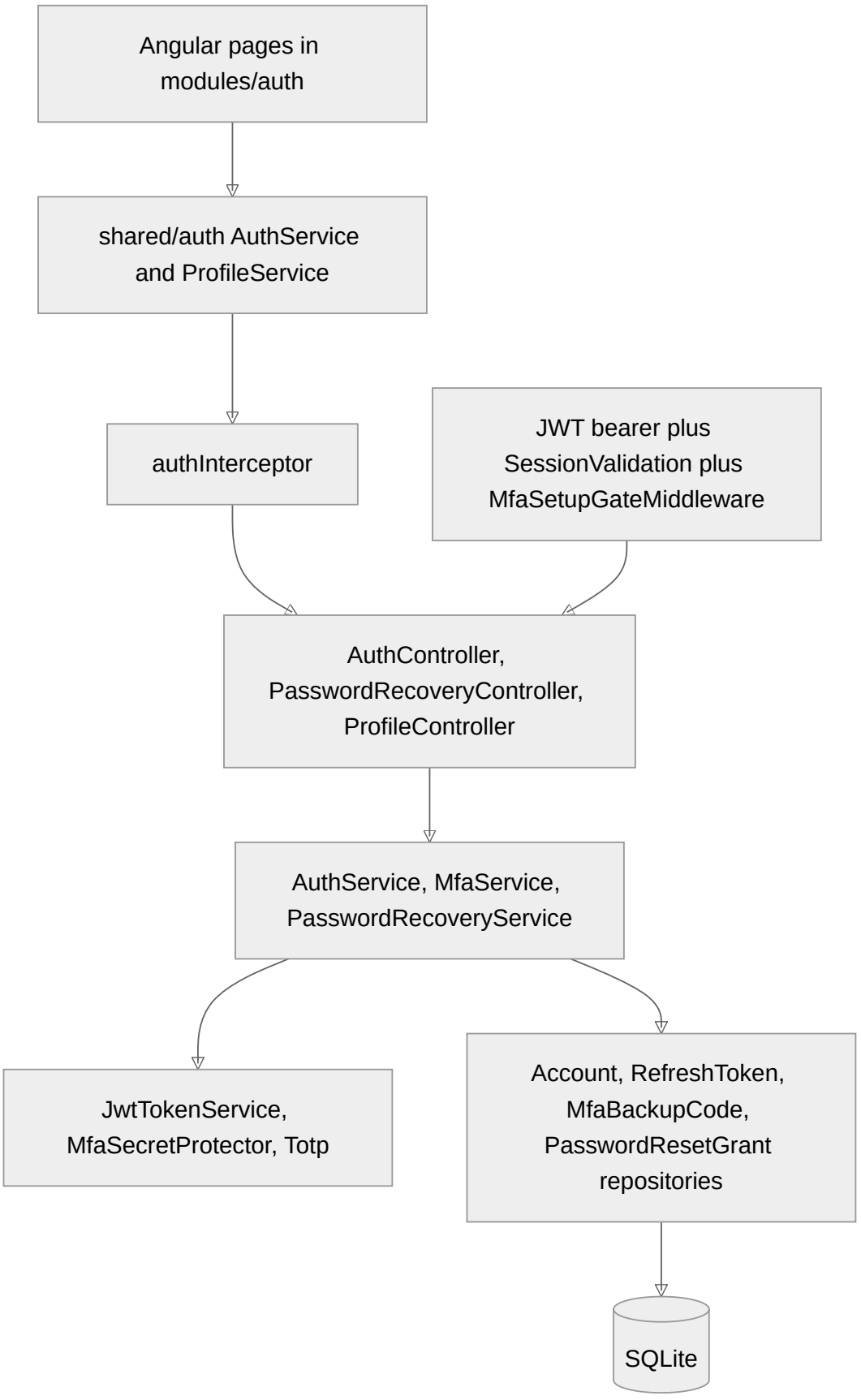
✅ Target framework confirmed: `net10.0` (.NET 10), from `backend/src/TaskFlow.Api/TaskFlow.Api.csproj`.

1. Technical Overview

The Authentication module issues and validates sessions for the TaskFlow SPA. It is built from:

- **Password sign-in** (`AuthController` → `AuthService`) using ASP.NET Core Identity's `PasswordHasher<User>` (PBKDF2), returning a short-lived **JWT access token** (HMAC-SHA256) in the body and a **refresh token** in an HttpOnly cookie (`taskflow_refresh`, path `/api/v1/auth`). Detail: `01.1-Login.md`, `01.4-Sessions-and-Tokens.md`.
- **Authenticator-app MFA (TOTP, RFC 6238)** implemented in-house (`Integrations/Security/OneTimeCodes/Totp.cs`, `Base32.cs`). The shared key is encrypted with ASP.NET Core Data Protection (`MfaSecretProtector`); the second sign-in step uses a Data-Protection-signed **challenge token** bound to the user id and security stamp. Ten single-use backup codes are stored as SHA-256 hashes. Detail: `01.3-MFA-Security-Setup.md`.
- **Mandatory MFA enrolment**: `MfaSetupGateMiddleware` returns `403` with code `mfa_setup_required` for every API call except the setup allow-list while `Mfa:Required` is true and the user has no MFA.
- **Account recovery** (`PasswordRecoveryController` → `PasswordRecoveryService`): an authenticator code (no backup code) buys a single-use, 10-minute reset token (`PasswordResetGrants` table); an in-memory sliding-window `AttemptLimiter` throttles per account and per caller address. Detail: `01.2-Account-Recovery.md`.
- **Session validation** (`Middleware/SessionValidation.cs`, hooked to `JwtBearerEvents.OnTokenValidated`): on every request the user's security stamp, active flag, role and effective permissions are re-read through `SessionService`, so password changes, resets, MFA resets, deactivation and role edits take effect on the next request.
- **Angular**: `modules/auth` (login, account-recovery, security-setup pages) plus `shared/auth` (`AuthService`, `authInterceptor`, guards, `TokenStore`, `role-routes.ts`). The access token is kept in memory only (`TokenStore` signal); silent refresh happens on `401` and at app start (`provideAppInitializer` → `restoreSession()`).

2. Architecture Flow



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3. Backend File Structure

All paths are under `TaskFlow Proj/backend/src/TaskFlow.Api/`. Each child document lists its own files; this table lists the folders and the files shared by several children.

Layer	Path	Files / owner
Controllers	`Controllers/Auth/`, `Controllers/Profile/`, `Controllers/Management/`	`AuthController.cs` (01.1, 01.4), `PasswordRecoveryController.cs` (01.2), `ProfileController.cs` MFA actions and `EmployeesController.cs` `ResetMfa` (01.3)
Services	`Services/Auth/{Login, Mfa, PasswordRecovery, Sessions, AttemptLimits}/`, `Services/Profile/`	`AuthService` (01.1, 01.4), `MfaService` (01.3), `PasswordRecoveryService` (01.2), `SessionService` (01.4), `ProfileService` MFA wrapper (01.3)
Service – Attempt limits (shared by 01.2 and 01.3)	`Services/Auth/AttemptLimits/`	`IAttemptLimiter.cs`, `AttemptLimiter.cs`
Repositories	`Repositories/Auth/{Accounts, RefreshTokens, Mfa, PasswordRecovery}/`	`AccountRepository` (all children), `RefreshTokenRepository` (01.4 owner; also 01.1, 01.2, 01.3), `MfaBackupCodeRepository` (01.3), `PasswordResetGrantRepository` (01.2)
DTOs (shared)	`Dtos/Auth/`	`UserDto.cs`, `AuthResponse.cs` (`AuthResponse`, `AuthenticatedUserResponse`), `AuthTokens.cs` (used by 01.1 and 01.4)
DTOs / validators per child	`Dtos/Auth/`, `Dtos/Auth/Mfa/`, `Dtos/Auth/PasswordRecovery/` and their `Validators/` folders	See each child's section 3
Integrations	`Integrations/Security/{Tokens, MfaSecrets, OneTimeCodes}/`	Tokens (01.4, used by 01.1), MfaSecrets (01.3, challenge used by 01.1), OneTimeCodes (01.3)
Entities (shared)	`Models/Entities/`	`User.cs`, `Role.cs`; per child: `RefreshToken.cs` (01.4), `MfaBackupCode.cs` (01.3), `PasswordResetGrant.cs` (01.2)
Enums	`Models/Enums/`	`NotificationType` (System) is used for notices (01.2, 01.3); `MfaVerifyOutcome` (01.3, used by 01.1), `RecoveryVerifyStatus` (01.2)
Common / Security (shared)	`Common/Security/`	`PermissionCatalog.cs`, `PermissionKeys.cs`, `PermissionPolicy.cs`, `SystemRoles.cs`; `AuthClaims.cs`, `RefreshTokenCookie.cs`, `EffectivePermissions.cs` (01.4, also used by 01.1)

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Layer	Path	Files / owner
Middleware	`Middleware/`	`GlobalExceptionHandler.cs` (shared); `SessionValidation.cs` (01.4); `MfaSetupGateMiddleware.cs` (01.3)
Filters (shared)	`Filters/`	`ValidationFilter.cs` (runs FluentValidation validators)
Authorization (shared)	`Authorization/Policies/`, `Authorization/Auditing/`	`RequirePermissionAttribute.cs`, `DeniedActionAuditHandler.cs`
Config	`Config/`	Shared: `IntegrationRegistration.cs` (token service, password hasher, Data Protection), `ServiceRegistration.cs`, `RepositoryRegistration.cs`; `JwtOptions.cs`, `CrossCuttingRegistration.cs` (01.4); `MfaOptions.cs` (01.3)
Persistence (shared)	`Persistence/`	`AppDbContext.cs`; migrations `20261011171611_InitialAuth.cs` (creates `Users`, `RefreshTokens`), `20261012173144_AddUserSecurityStamp.cs` (adds `Users.SecurityStamp`), `20261023101938_AddMfaAndAccountRecovery.cs` (adds the `Users.Mfa*` columns and `MobileNumber`, creates `MfaBackupCodes`, `PasswordResetGrants`)

4. Frontend File Structure

All paths are under `TaskFlow Proj/frontend/src/app/`. Child-specific files are listed in each child document.

Type	Path	Purpose
Routes	`modules/auth/auth.routes.ts`	`AUTH_ROUTES`: `login` (01.1), `account-recovery` (01.2), `security-setup` (01.3), all outside the shell
Page components	`modules/auth/login/`, `modules/auth/account-recovery/`, `modules/auth/security-setup/`	One folder per screen (01.1, 01.2, 01.3)
API service (shared)	`shared/auth/auth.service.ts`	`login`, `verifyMfa` (01.1); `verifyAuthenticatorRecovery`, `recoverPassword` (01.2); `refresh`, `restoreSession`, `logout`, `checkSession` (01.4)
Models (shared)	`shared/auth/auth.models.ts`	`AuthUser`, `LoginRequest`, `AuthResponse`, `MfaChallenge`, `RecoveryTokenResponse`, `RecoverPasswordRequest`, `MfaSetup`, `MfaBackupCodes`, `isMfaChallenge()`
Route / Guard (shared)	`shared/auth/auth.guard.ts`	`authGuard` (01.4), `mfaSetupGuard`, `securitySetupGuard` (01.3), `permissionGuard` (used by other modules)
Interceptor	`shared/auth/auth.interceptor.ts`	01.4 (refresh on `401`) and 01.3 (`403 mfa_setup_required`)

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Type	Path	Purpose
Messages (shared)	`shared/auth/login-reasons.ts`	`reason` query values and login notices (01.1, 01.4)
Shell / bootstrap	`layouts/shell.component.ts`, `config/app.config.ts`, `routes/app.routes.ts`	01.4

5. Child Modules

#	Child Module	Main endpoints (count)	Key permissions	Document
01.1	Login	`POST /api/v1/auth/login`, `POST /api/v1/auth/mfa/verify`, `GET /api/v1/auth/me` (3)	None (`[AllowAnonymous]`; `/auth/me` `[Authorize]`)	`01.1-Login.md`
01.2	Account Recovery	`POST /api/v1/auth/password-recovery/authenticator/verify`, `POST /api/v1/auth/password-recovery/reset` (2)	None (`[AllowAnonymous]`)	`01.2-Account-Recovery.md`
01.3	MFA Security Setup	`POST /api/v1/users/me/mfa/setup`, `/enable`, `/check`, `/backup-codes`, `POST /api/v1/users/{id}/mfa/reset` (5)	`profile.view`, `profile.edit`, `profile.mfa.manage` (Always On), `employees.mfa.reset`	`01.3-MFA-Security-Setup.md`
01.4	Sessions and Tokens	`POST /api/v1/auth/refresh`, `POST /api/v1/auth/logout` (2)	None (`[AllowAnonymous]`, cookie)	`01.4-Sessions-and-Tokens.md`

Endpoint numbers 1–12 are kept across the child documents so they can be cross-referenced. The child documents are in the folder `01-Auth/` next to this file.

6. Database Design

SQLite types come from the migrations; table names are the `DbSet` names in `AppDbContext`. `Users` is shared by every child and is described in full here; the other tables are described in the child that owns them.

Table	Owner child	Used by
`Users`	Shared (module)	01.1 (sign-in, `LastLoginAt`), 01.2 (password, stamp), 01.3 (`Mfa*` columns, stamp), 01.4 (stamp, active flag)
`RefreshTokens`	01.4 Sessions and Tokens	01.1 (insert on sign-in), 01.2 and 01.3 (revoke all on reset)
`MfaBackupCodes`	01.3 MFA Security Setup	01.1 (backup code at sign-in)
`PasswordResetGrants`	01.2 Account Recovery	–

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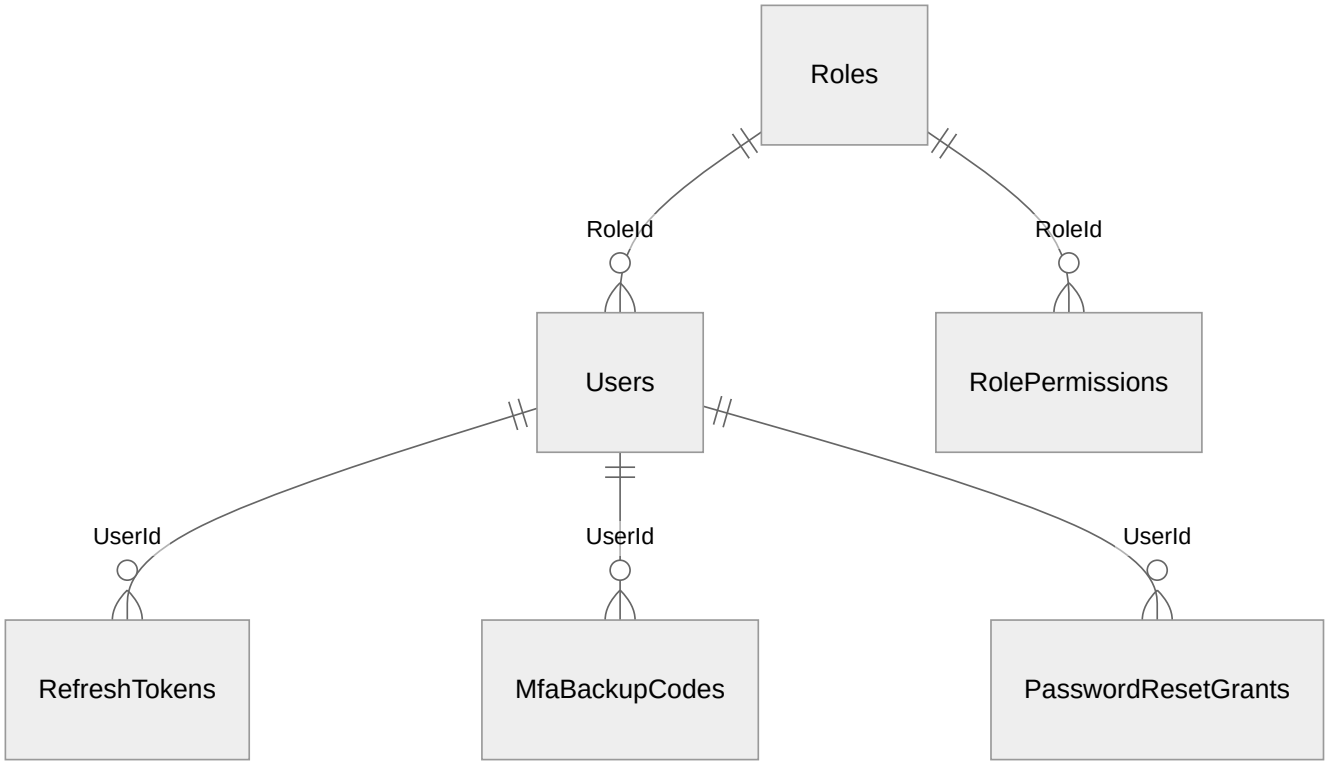
Table	Owner child	Used by
`Roles`, `RolePermissions`, `Permissions`	Settings module	01.1, 01.4 (role and permission keys read)

Table	Column	Type	Constraints	Description
`Users`	`Id`	INTEGER	PK, autoincrement	User id
`Users`	`EmployeeCode`	TEXT(32)	Nullable	Employee code
`Users`	`FullName`	TEXT(200)	Not Null	Display name
`Users`	`Username`	TEXT(100)	Not Null, Unique (`IX_Users_Username`)	Sign-in name
`Users`	`Email`	TEXT(256)	Not Null, Unique (`IX_Users_Email`)	Sign-in email (stored lower-case by `EmailPolicy.Normalize`)
`Users`	`MobileNumber`	TEXT(20)	Nullable	E.164 number (collected on Security Setup)
`Users`	`PasswordHash`	TEXT	Not Null	`PasswordHasher<User>` hash
`Users`	`SecurityStamp`	TEXT(64)	Not Null	Rotated on password change/reset and MFA reset; compared with JWT `sstamp`
`Users`	`MfaSecretProtected`	TEXT(512)	Nullable	Data-Protection-encrypted TOTP key
`Users`	`MfaEnabledAt`	TEXT (DateTime)	Nullable	When MFA was turned on
`Users`	`MfaLastUsedStep`	INTEGER (long)	Not Null, concurrency token	Last accepted TOTP step (replay protection)
`Users`	`MfaFailedAttempts`	INTEGER	Not Null, default 0, concurrency token	Consecutive wrong codes
`Users`	`MfaLockedUntil`	TEXT (DateTime)	Nullable	MFA lockout end
`Users`	`RoleId`	INTEGER	FK → `Roles.Id`	Role
`Users`	`IsActive`	INTEGER (bool)	Not Null	Inactive users cannot sign in or keep a session
`Users`	`LastLoginAt`	TEXT (DateTime)	Nullable	Set on successful sign-in
`Users`	`CreatedAt`, `UpdatedAt`	TEXT (DateTime)	Not Null	Audit times
`Users`	`IsDeleted`	INTEGER (bool)	Not Null; global query filter `!IsDeleted`	Soft delete

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Relationships

- `Users.RoleId → Roles.Id` (FK, Restrict)
- `RefreshTokens.UserId → Users.Id` (FK, Cascade); query filter hides tokens of deleted users
- `MfaBackupCodes.UserId → Users.Id` (FK, Cascade)
- `PasswordResetGrants.UserId → Users.Id` (FK, Cascade)
- Permissions are read through `RolePermissions.RoleId → Roles.Id` and `RolePermissions → Permissions.Key` (Settings module).



7. Security & Authorization

- **Authentication:** JWT bearer (`Microsoft.AspNetCore.Authentication.JwtBearer`), access token 30 min in memory on the client; refresh token 7 days in cookie `taskflow_refresh` (HttpOnly, Secure, SameSite=Strict, Path `/api/v1/auth`), rotated on every refresh and stored only as a SHA-256 hash (detail in 01.4).
- **Pipeline order (`Program.cs`):** `UseCors` → `UseAuthentication` → `MfaSetupGateMiddleware` → `UseAuthorization`.
- **Session revocation:** security stamp check on every request (`SessionValidation`); role and permission claims re-read from the database every request.
- **Anti-enumeration:** login returns a bare `401` for any failure; recovery returns one message for every failure.
- **Authorization:** permission policies `perm:{key}` are generated for every `PermissionCatalog.All` entry (`RequireClaim("permission", key)`); denied authorizations are logged by `DeniedActionAuditHandler` ("Denied action: user … attempted …").
- **Data scope:** Own – every `/users/me` MFA endpoint takes the user id from the token (`ClaimTypes.NameIdentifier`), never from the request. Organization – `employees.mfa.reset` applies to any user id.

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- **Shared attempt limiter:** `AttemptLimiter` (singleton, in memory) – sliding 5-minute window, `AttemptsPerAccount = 5`, `AttemptsPerAddress = 20`, `MaxTrackedKeys = 10 000` (oldest evicted), idle keys swept every window; IPv6 addresses pooled by /64, IPv4-mapped addresses normalised. Used by recovery (per account + address, 01.2) and by password re-entry for MFA setup/backup-code regeneration (key `pwd:{id}`, no address, 01.3).
- **Frontend:** `authGuard` (redirects to `/login?returnUrl=`), `mfaSetupGuard` on the shell, `securitySetupGuard` on `/security-setup`; the **Reset MFA** button in Employee Management is shown only with `employees.mfa.reset` and when the employee has MFA on.

Permission Key	Endpoint(s)	UI Element Controlled
(none – `[AllowAnonymous]`)	`POST /api/v1/auth/login`, `/auth/mfa/verify`, `/auth/refresh`, `/auth/logout`, `/auth/password-recovery/authenticator/verify`, `/auth/password-recovery/reset`	Login screen, Account Recovery screen, Sign out button
(none – `[Authorize]`)	`GET /api/v1/auth/me`	Session check, permission sync
`profile.view` (Always On, not attribute-checked)	`GET /api/v1/users/me`	Security Setup step 1 facts
`profile.edit` (Always On, not attribute-checked)	`PUT /api/v1/users/me`	Security Setup step 1 form
`profile.mfa.manage` (Always On, not attribute-checked)	`POST /api/v1/users/me/mfa/setup`, `/enable`, `/check`, `/backup-codes`	Security Setup steps 2–3
`employees.mfa.reset` (Administrator Granted; Manager/Employee Denied; Custom Configurable)	`POST /api/v1/users/{id}/mfa/reset`	**Reset MFA** button in the employee details panel

8. Validations & Error Handling

Shared rules only; child-specific rules are in each child document.

Rule	Layer	HTTP Code	Error Message
MFA locked	`AuthController` / `ProfileController.CheckMfa` / `MfaService`	429 / 400	Too many wrong codes. Try again in a few minutes. (429 from controllers; 400 when thrown as `BusinessRuleException` from setup/enable/backup-codes)
Parallel save conflict	`GlobalExceptionHandler` (`DbUpdateConcurrencyException`, `ConcurrencyConflictException`)	409	Someone else changed this record while you were saving. Reload it and try again.
Unhandled exception	`GlobalExceptionHandler`	500	Something went wrong. Please try again.

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Request DTOs are validated by FluentValidation through `ValidationFilter` (validation ProblemDetails `errors` map). Errors are returned by the global exception middleware (`GlobalExceptionHandler`) and by controller `Problem(...)` calls as `application/problem+json`. `RateLimitedException` can add a `Retry-After` header when a delay is supplied (none is supplied in this module).

9. Integrations / Side Effects

- **Notifications:** `INotificationPublisher.PublishAsync(..., NotificationType.System, ...)` after a password recovery reset (01.2) and after an administrator MFA reset (01.3); texts are in the child documents.
- **Email:** none. No password, code or link is sent by email or SMS in this module.
- **Activity Log:** no task activity entries. Denied authorizations are written to the application log by `DeniedActionAuditHandler`; recovery notification failures are logged as warnings (01.2).
- **Session effects:** password reset and admin MFA reset rotate `SecurityStamp` and revoke all refresh tokens (all sessions end).
- **AI Assistant:** > ⚠️ To be confirmed: no assistant tool for this module was found in `Controllers/Auth` or `Services/Auth`; the assistant's help knowledge has `change-password.md` (Profile) but no sign-in or MFA topic.

10. Configuration

Shared keys only; `Jwt:*` keys are in 01.4, `Mfa:Issuer`, `Mfa:BackupCodeCount` in 01.3, `Mfa:ChallengeMinutes` in 01.1.

Key (`appsettings.json`)	Purpose	Default
`Mfa:Required`	Force MFA enrolment for everyone (break-glass: `Mfa__Required=false`); used by sign-in (`mfaSetupRequired`), session validation and the setup gate	`true`
`Mfa:MaxFailedAttempts`	Wrong codes before lockout (code default, not in appsettings); shared by sign-in, code check and backup-code regeneration	`5`
`Mfa:LockoutMinutes`	Lockout length (code default)	`5`
`AiProviders:KeyRingPath`	Data Protection key ring folder (also protects MFA keys and sign-in challenges)	`dp-keys`
`Storage:Root`	Base folder for the key ring and database	`""`
`ConnectionStrings:Default`	SQLite database	`<secret>`

Hard-coded shared values: `AttemptLimiter` 5 per account, 20 per address, 5-minute window, 10 000 keys.

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11. Testing

Paths are relative to `backend/tests`. Child-specific tests are listed in each child document; the files below cover several children.

Test Type	File / Location	Covers
Unit (support)	`TaskFlow.UnitTests/Auth/Login/AuthServiceHarness.cs`	Test harness for `AuthService` (01.1, 01.4)
Unit (support)	`TaskFlow.UnitTests/TestSupport/AcceptingMfaService.cs`	MFA stub
Unit	`TaskFlow.UnitTests/Auth/AttemptLimits/AttemptLimiterTests.cs`	Sliding window limiter (01.2, 01.3)
Integration	`TaskFlow.IntegrationTests/Auth/Login/AuthEndpointsTests.cs`	Login, refresh, logout, me endpoints (01.1, 01.4)
Integration	`TaskFlow.IntegrationTests/Common/PermissionAuthorizationTests.cs`	Permission policies
Integration (support)	`TaskFlow.IntegrationTests/TestSupport/MfaTestSupport.cs`	MFA helpers

No frontend spec files were found under `modules/auth`.

12. Known Limitations / Future Enhancements

- `AttemptLimiter` is in memory: a restart resets it and it is not shared between multiple API instances.
- **TDD vs code:** `Artifacts/TDD-TaskFlow.md` §8.2 lists `POST /api/v1/auth/change-password` and logout as "Authenticated"; in code password change is `POST /api/v1/users/me/password` (Profile) and logout is `[AllowAnonymous]` and cookie-based. MFA, recovery and the setup gate are not described in the TDD.
- **Permission audit vs code:** `permission-inventory.csv` cites `Controllers/MfaController.cs`, `features/auth/...`, `core/auth/...` and `features/onboarding/...`; the code has no `MfaController` (reset is `EmployeesController.ResetMfa`, self-service is `ProfileController`) and the frontend lives in `modules/auth` and `shared/auth`.
- **Permission audit finding is out of date:** permission-gaps.md B.3 says `UsersController.Update` sets a `Password` field without an authenticator code. In the current code `UpdateUserRequest.Password` is marked "Ignored" and `EmployeeService` hashes a password only in `CreateAsync`, so the bypass is no longer present.

Child-specific gaps are listed in each child document.

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